

Molecular benchmark report: LiH, BeH₂, HF, and H₂O through `clifford_qc` A-CASE

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1 Results

Errors in mHa against the sector-exact ground energy of the reference’s own (N, S_z) sector; chemical accuracy is 1.5936 mHa.

Table 1: Adaptive A-CASE against classical baselines.

Molecule	n	N_e	Sector dim	M	A-CASE (mHa)	CCSD (mHa)	CISD (mHa)	SD (mHa)	$\kappa(S)$
BeH ₂	14	6	1225	11	+1.3411	+0.3934	+0.7526	+0.7526	1.00
BeH ₂ 2×	14	6	1225	31	+28.5144	+5.8613	+28.5110	—	1.00
H ₂ O 2.5×	14	10	441	21	+10.0927	−41.1309	+10.0926	—	1.00
H ₂ O	14	10	441	21	+1.4477	+0.1165	+0.7051	+0.7051	1.00
H ₂ O 2×	14	10	441	21	+55.4206	−9.7177	+54.5703	—	1.00
HF	12	10	36	5	+1.0725	+0.0000	+0.0000	+0.0000 [†]	1.00
LiH	12	4	225	6	+1.1264	+0.0105	+0.0133	+0.0133	1.00
LiH 3×	12	4	225	5	+1.5210	+0.1320	+0.1509	—	1.00

Reading the table. A-CASE reaches chemical accuracy on all 8 molecules. CCSD is more accurate on 7 of 8 (BeH₂, BeH₂ 2×, H₂O, H₂O 2×, HF, LiH, LiH 3×) at a small fraction of the cost. The adaptive subspace is not competitive with classical coupled cluster on these systems and is not presented as if it were; what it shows is that the operator-response subspace clears the threshold at small M against a sector two to three orders larger.

The stopping rule is oracle-assisted. Growth halts on the first step whose error against the *exact* energy falls below threshold, so the M column answers how small the basis can be while clearing the bar — a property of the selector, not a cost the method could reproduce without knowing the answer.

[†] The complete SD basis spans the entire sector for this molecule, so agreement with exact diagonalization is arithmetic rather than accuracy.

2 Observables

Read off the same solve that produced the adaptive energy, through the projected-observable route; no Ritz state is materialized.

Table 2: Double occupancy against its uncorrelated value, and the singlet check.

Molecule	$\langle d \rangle$	uncorrelated	difference	$\langle S^2 \rangle$
BeH ₂	0.426218803	0.428571429	$-2.35e-03$	$2.06e-04$
BeH ₂ 2 $\times$	0.395619366	0.428571429	$-3.30e-02$	$8.73e-18$
H ₂ O 2.5 $\times$	0.428571429	0.714285714	$-2.86e-01$	$6.00e+00$
H ₂ O	0.711243361	0.714285714	$-3.04e-03$	$3.81e-04$
H ₂ O 2 $\times$	0.656131728	0.714285714	$-5.82e-02$	$2.66e-01$
HF	0.833034591	0.833333333	$-2.99e-04$	$3.01e-34$
LiH	0.332419497	0.333333333	$-9.14e-04$	$1.23e-32$
LiH 3 $\times$	0.325510212	0.333333333	$-7.82e-03$	$1.86e-02$

The S_z Lehmann response is zero on every molecule: S_z couples a singlet to triplets, and the singlet-preserving excitation basis spans none, so χ_0 here is a property of the basis rather than of the molecule and no response result is claimed.

3 Resource accounting

Table 3: Distinct Pauli words in H against the A-CASE element word universe $|\bigcup_{ij} \text{supp}(A_i^\dagger A_j) \cup \text{supp}(A_i^\dagger H A_j)|$. The two are different quantities.

Molecule	Words in H	Element universe	M	SD wall-clock (s)
BeH ₂	666	350,196	11	3018.3
BeH ₂ 2 $\times$	666	1,165,663	31	—
H ₂ O 2.5 $\times$	1086	994,052	21	—
H ₂ O	1086	1,168,272	21	2461.3
H ₂ O 2 $\times$	1086	874,597	21	—
HF	631	21,156	5	54.2
LiH	631	38,250	6	440.3
LiH 3 $\times$	631	24,096	5	—